

Yu Sheng Aow

312-804-7949 | ays03@bu.edu | linkedin.com/in/yushengaow | github.com/yusheng036

EDUCATION

Boston University

Bachelor of Arts in Computer Science; Dean's List (Fall 2025)

Boston, MA

Sep. 2024 – Present

University of Illinois Chicago

Bachelor of Science in Computer Science

Chicago, IL

Aug. 2023 – Aug. 2024

EXPERIENCE

Undergraduate Research Assistant

COMMIT Compiler Group, MIT CSAIL, & Quansight Consulting

Jul. 2025 – Present

Cambridge, MA

- **Sparse Tensor Statistics Suite:**

- * **Feature Design:** Proposed a technical design for statistics estimation to support query optimization in a sparse tensor compiler, defining a new statistics interface to improve estimator accuracy and runtime efficiency, fielding team design reviews with faculty advisors.
 - * **Feature Implementation:** Developed prototype sparse tensor estimators using structured matrix multiplication and graph-based sampling to improve estimation accuracy for 2D and 3D tensor operations.
 - * **Feature Delivery:** Constructed a benchmark infrastructure to evaluate estimator accuracy and runtime across established sparse tensor benchmarks, producing evaluation reports and publication-ready visualizations.
- **Pessimistic Cardinality Estimator:** Designed and executed a pessimistic cardinality estimator leveraging chain bounds and breadth-first search to generate efficient query plans in the relational setting.
 - **STC Kernel Execution:** Built a cost-based logical optimizer leveraging greedy search to cultivate synchronized execution plans for sparse tensor contractions and delivered a stable plan that avoided timeouts on benchmarks where modern query engines like DuckDB and Umbra failed.
 - **Tensor Interface Integration:** Implemented eager and lazy tensor interfaces in the Finch sparse tensor compiler, extending functionality while ensuring compliance with the Python Array API standard and maintaining backward compatibility across existing tensor operations.

GitHub Repository: [finch-tensor-lite](#)

Undergraduate Research Assistant

BU AI Research Initiative (AIR) / Image Video Computing (IVC)

Jan. 2026 – Present

Boston, MA

- **Distributed LoRA Fine-Tuning:** Trained 8B-parameter vision-language models using a streamlined fine-tuning pipeline on Modal. Assembled a custom GPU environment with CUDA and video-processing libraries. Orchestrated a multi-GPU training, checkpointing, and cloud storage to enable reliable large-scale model development.
- **Multi-Model Evaluation Framework:** Developed a benchmarking pipeline for vision-language models, executing batched inference across architectures, computing standardized evaluation metrics, and aggregating results into reproducible reports to lowering manual evaluation overhead and enable consistent comparison.
- **Data Annotation Collection:** Deployed a video annotation workflow incorporating Label Studio to design systematized prompts and reasoning-focused questions, achieving high-quality training datasets for vision-language model fine-tuning while lowering labeling noise.

PROJECTS

Localized Region Bound (LRB) Estimator | *Python, Scipy, Matplotlib*

Jan. 2026 – Present

- **LRB Formulation:** Designed and engineered a locality-aware pessimistic nnz estimator for n-dimensional sparse tensors, analyzing reduced-axis sparsity arrangement to support multi-stage matrix multiplication chains.
- **Sparse Matrix Benchmark Suite:** Built synthetic and SuiteSparse benchmarking frameworks to evaluate estimator performance across edge-case and real-world sparse matrix workloads.
- **Performance Evaluation:** Demonstrated LRB on 2D and 3D SpMM outputs across synthetic and SuiteSparse workloads, reducing overestimation from over 10× to near 1–2× of true nnz as region granularity increased.

GitHub Repository: [sparse-tensor-stats-suite](#)

TECHNICAL SKILLS

Languages: Java, Python, C, C++, Assembly, Julia

Frameworks: PyTorch, DeepSpeed, Flask, FastAPI, React, Node.js

Misc: NumPy, SciPy, pandas, Matplotlib, CUDA, Modal, Git, Linux, SuiteSparse